

**Name - Surname:** Naruebest Komol

**Age:** 28 years

**Position:** Electronic Engineer

**Affiliation:** Centre of Observatory Operations and Engineering (COOE)

**Education :** Bachelor's Degree in Electrical and Electronics Engineering, Rajamangala University of Technology Lanna

**Work Experience :**

- Engineer (July 8, 2020-Present) National Astronomical Research Institute of Thailand (NARIT)

**Technical Knowledge & Skills:**

PCB Design & Hardware:

Designing and analyzing electronic circuits.

PCB schematic Design & Layout (Altium, Eagle, EasyEDA, KiCad, PCB Wizard)

PCB fabrication and component soldering/assembly following IPC PCB standards

Reading, analyzing, and designing electronic schematics

Basic Understanding of EMC/EMI design considerations

Creating Wiring diagrams using Visio

General embedded/hardware troubleshooting and system design

Embedded Systems & Programming:

Programmable Logic Controllers (PLC- Mitsubishi, Omron) and automation system design

Microcontrollers (STM32, ESP32, Arduino)

Low-level driver development and peripheral configuration (UART, SPI, I2C, CAN, etc.)

Programming in C for embedded systems

Mechanical & Design Tools:

Mechanical design using SolidWorks, AutoCAD

Convert 3D designs into physical parts and verify mechanical compatibility

Testing & Measurement

Oscilloscope / Multimeter / Power Analyzer Usage

Circuit Simulation (PSpice, Proteus, Multisim)

Hardware Debugging and Troubleshooting

Documentation & Office Tools:

Document control and technical record management

Proficient in Microsoft Office Suite (Word, Excel, PowerPoint)

Prepare user manuals, reports, and project documentation

**Soft Skills:**

Communication & Coordination: Effectively communicate within teams and with other departments.

Writing & Presentation: Prepare reports, documentation, and present technical or project information clearly.

Problem Solving: Analyze and resolve issues quickly and appropriately in various situations.

**Physical Qualifications / Special Conditions:**

Good health, able to travel for assignments domestically and internationally.

Holder of a valid driver's license.

**Other Information:**

Certifications (IPC PCB Soldering Certificate, Altium designer advance course, SolidWorks Essentials course, SolidWorks Simulation course, SolidWorks Assembly course)

**Summary of Responsibilities (Engineer):**

1. Design, develop, and test systems/equipment according to project requirements.
2. Install, inspect, and maintain engineering systems and instruments.
3. Prepare technical documents and performance reports.
4. Coordinate with research teams and relevant departments.
5. Perform other assigned tasks as required.

**Details of work experience :****1. Designed Electrical Integrated Control Cabinets at all observatories of the institute.**

Designed the complete system architecture, PLC programs, HMI interfaces, and equipment layout.

It is an integrated electrical control cabinet for the entire observatory. The cabinet consists of a computer, a network switch, and a PLC-based control system for dome operations. It is equipped with an HMI interface for local control and also supports remote operation via the Thai Robotic Telescope (TRT). The installation history is as follows:

- (2020) 1-meter Telescope at Thai National Observatory (TNO)
- (2020) Project DIMM at Thai National Observatory (TNO)
- (2020) Electrical Integrated Control Cabinet at Regional Observatory for the Public, Songkhla (ROP-SKA)
- (2021) Electrical Integrated Control Cabinet at Regional Observatory for the Public, Chachoengsao (ROP-CCO)
- (2021) 12-foot Telescope Control Cabinet at Regional Observatory for the Public, Songkhla (ROP-SKA)
- (2022) Electrical Integrated Control Cabinet at AstroPark Observatory (APK)
- (2022) Electrical Integrated Control Cabinet at Spring Brook Observatory (SBO)
- (2023) Electrical Integrated Control Cabinet at Cerro Tololo Inter-American Observatory (CTIO), Chile
- (2023) Electrical Integrated Control Cabinet at Regional Observatory for the Public, Khon Kaen (ROP-KKN)
- (2024) Electrical Integrated Control Cabinet at Sierra Remote Observatories (SRO), USA
- (2024) Electrical Integrated Control Cabinet at Gao Mei Gu Observatory, Lijiang, China
- (2024) Sliding Roof Control Cabinet at Regional Observatory for the Public, Chachoengsao (ROP-CCO)



Figure 1 : Designed diagram of The control cabinet, containing PLC, computer, network switch, GPS, and other equipment, which has been installed at all observatories of the institute.

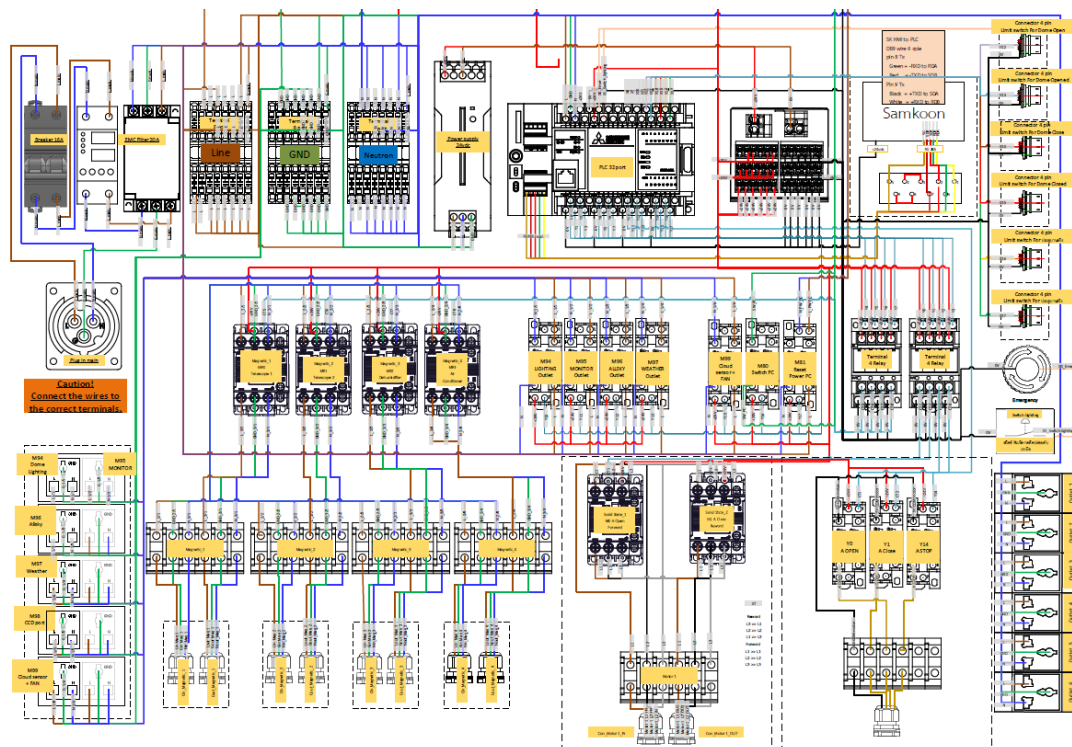


Figure 2 : Wiring Diagram the control cabinet

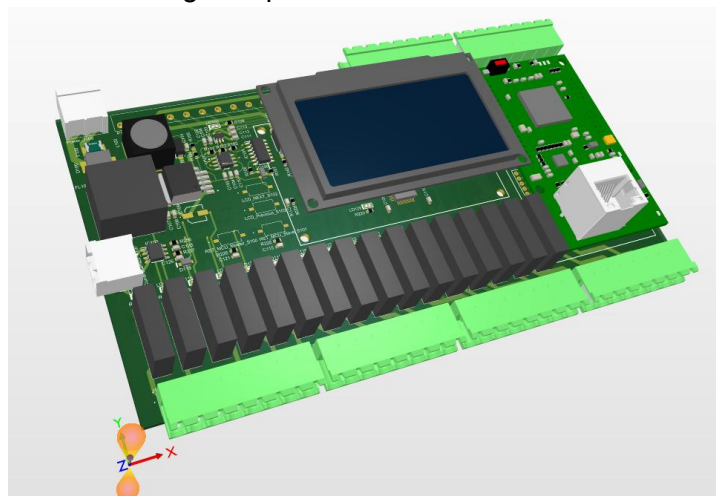


*Figure 3 : The control cabinet installation at Cerro Tololo Inter-American Observatory (CTIO), Chile, in 2023.*

## 2. PCB Schematic Design & Layout

Designed PCBs ranging from 2-layer to 8-layer boards, developing power distribution systems for on-board devices and microcontroller circuits. Considered EMC/EMI requirements in all PCB designs to ensure signal integrity and reliability. Performed soldering and assembly following IPC PCB standards, ensuring high-quality fabrication. Managed complete PCB projects from schematic design to final assembly. Successfully delivered multiple functional boards for embedded and control systems applications.

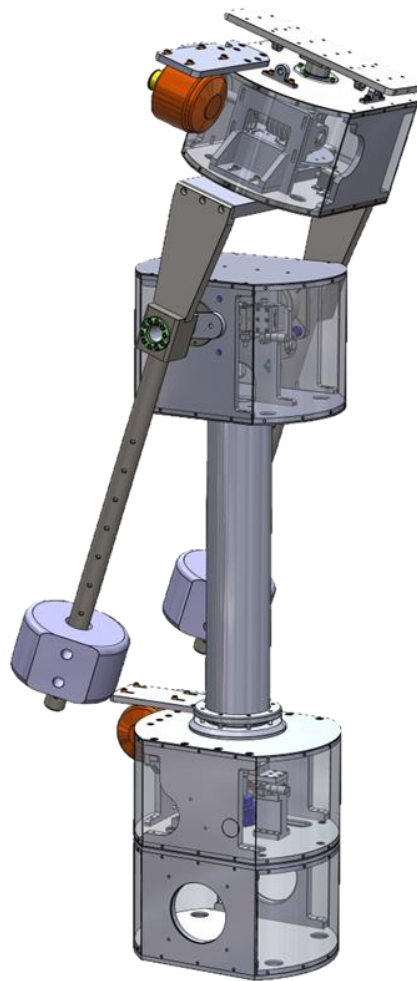
- Filter Wheel Controller PCB (v1, v2).
- Allsky Camera PCB (v1).
- Balloon PCB with ESP32 LoRa32.
- ESP32 Analog Read Temperature Universal Board
- ESP32 Power Metering Board.
- Smart PLC Using STM32
- ESP32 Analog Temperature Universal Board



*Figure 4 : Designed The Smart PLC Using STM32 With OLED interface and LAN Ethernet Board*

### 3. Designed the Radio Frequency Interference (RFI) Tracking Mount

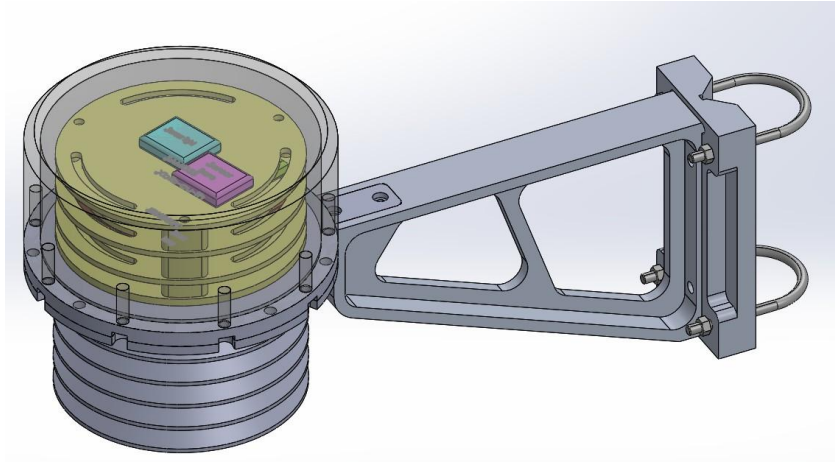
The Radio Frequency Interference (RFI) tracking mount is a specialized platform designed for the deployment of instruments used in detecting radio frequency interference. It features dual-axis rotational capabilities (Azimuth and Altitude), allowing the equipment to be precisely aimed in any direction for optimal signal monitoring. This mount also supports both sidereal and non-sidereal tracking, making it suitable for missions requiring accurate positioning of signal sources. In fiscal year 2024, the institution successfully procured the required materials and completed the fabrication of all mechanical components. The system is currently undergoing assembly and testing, with full operational readiness expected by the end of fiscal year 2025. This equipment plays a vital role in monitoring and analyzing environmental interference, thereby enhancing the precision of astronomical data collection and elevating the quality of scientific research within the country.



*Figure 5 : The Radio Frequency Interference (RFI)*

#### 4. Designed The Housing Weather Station (v1)

The Weather Station v1 is a prototype system designed to monitor essential environmental parameters such as temperature, humidity, pressure, and wind conditions. The system integrates multiple sensors with a microcontroller platform for real-time data acquisition. Data is transmitted via wireless communication for remote monitoring and logging. The design emphasizes low power consumption and modular expandability for future upgrades. This project demonstrates practical skills in sensor integration, PCB design, and embedded system development.



*Figure 5 : Designed the Housing for Weather Station (v1)*

#### 5. Designed Software and PCB for the TRT Precision Filter Wheel.

Designed and developed software, PCB and Interface for The TRT Precision Filter Wheel, which supports up to 23 filters in a single device. The system supports network control via LAN and is fully compatible with the Thai Robotic Telescope Network (TRT). It also enables automated remote filter switching, enhancing convenience and efficiency in astronomical observations.

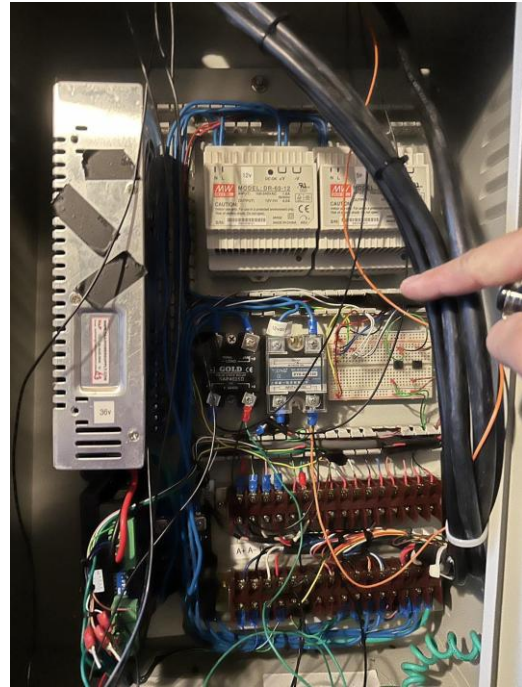


*Figure 6 : The TRT Precision Filter Wheel.*

## 6. Maintenance work



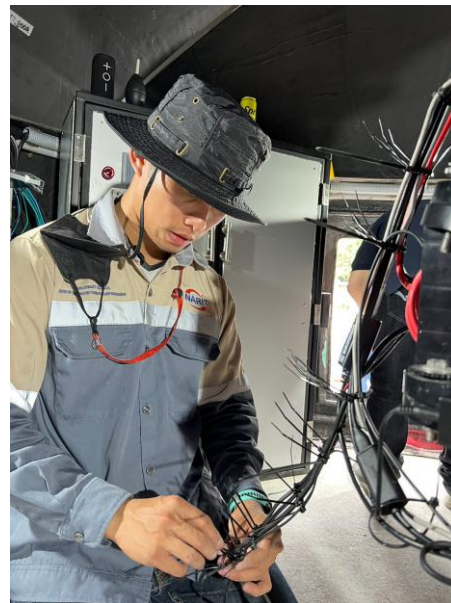
*Figure 7 : Annual maintenance of the telescope at Sierra Remote Observatory (SRO) in 2024*



*Figure 8 : Improved the drive system of the MRES equipment for greater stability and efficiency.*



*Figure 9 : Setup Weather Station at Astropark*



*Figure 10 : Installed all electrical systems within the dome at Regional Observatory for the Public. Sonakhla (ROP-SKA)*

## 7. Certificate



Certificate of Attendance  
awarded to  
Mr.Naruebest komol  
  
in recognition of successful participation in  
**Altium Designer Advanced course**  
  
During 10 June 2021 – 11 June 2021  
  
Boonchai Kingrunsped  
Marketing Manager  
Embedded Technology Co.,Ltd.

Figure 11 : Altium Designer Advance course Certificate



Figure 12 : Solidwork Essentials course Certificate



Figure 13 : Solidwork Assembly course Certificate



Figure 14 : Solidwork Simulation course Certificate



Figure 15 : IPC PCB Soldering course Certificate